EcoLoop: Honest Builder Recording Script

100% authentic, grounded, peer-to-peer read-aloud script for video recording.

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Target: 12 to 14 Minutes

> **Tone & Pacing:** Calm, natural, grounded. Talk like you're showing a project to other high school makers. Zero corporate jargon. The physical crafts and wrapper exchange were already happening; your job was designing the circular reinvestment loop, uniting it under EcoLoop, and building the digital storefront in one night using an AI agent.

SLIDE 01 • TITLE

EcoLoop: Circular material systems and zero-cost digital architecture

"Hey everyone. I'm Harman, from Swami Sant Dass Public School in Jalandhar, India."

"Look, most school environmental projects are kind of a joke. People make a poster, pick up a few plastic bottles after lunch, take a picture for social media, and call it sustainability. We wanted to actually build real things instead."

"So EcoLoop has two parts: first, taking waste that municipal recyclers literally throw away and turning it into solid materials. And second, building a digital platform to track those materials and take inquiries with zero dollars in running costs."

"Let's look at the packaging problem first." `[NEXT SLIDE]`

SLIDE 02 • THE PROBLEM

The Packaging Trap: Multi-Layer Plastics

"When people talk about recycling, everyone focuses on plastic water bottles. But bottles are actually easy to shred and melt down. The real nightmare is multi-layer packaging, or MLP: chips packets, chocolate wrappers, biscuit packaging."

"They laminate shiny aluminum foil between thin plastic sheets. Because those layers are chemically bonded, municipal recycling plants don't touch them. It costs more money to separate the foil than the plastic is worth. So virtually 100% of snack packaging ends up burned or dumped in a landfill."

"Meanwhile, every school canteen generates sacks of clean wrappers every single day. So our solution wasn't separating the layers; it was heat-compressing the whole thing together into heavy structural sheets that can replace commercial plywood." `[NEXT SLIDE]`

`SLIDE 03 • PHYSICAL PRODUCTS`

The Physical Loop

"Here's the honest breakdown of our physical product loop."

"Inside school, students actually make five products ourselves: handmade Seed Paper with plantable seeds, citrus Bioenzymes fermented from fruit peels, Uniform Bags stitched from old school uniform fabrics, Bottle Planters for gardening, and Eco-Bricks packed with plastic wrappers."

"Now, for the MLP Sheets on the bottom right: you need industrial hydraulic heat presses to compress wrappers, which obviously a school lab doesn't have. We don't pretend we manufacture those in class. Instead, there's a local NGO that takes clean snack wrappers in bulk by weight, requiring a fixed minimum batch, and returns finished, compressed MLP sheets. Those boards are rigid, waterproof, and can be sawed or drilled just like plywood. By linking our school collection to their bulk processing, we get structural boards out of clean canteen waste." `[NEXT SLIDE]`

`SLIDE 04 • HOW IT STARTED VS HOW IT'S GOING`

how it started vs how it's going

"When we started, all the individual pieces were already lying around the school: students were making crafts across different classrooms, and the school was already collecting snack wrappers by the kilo to exchange for MLP sheets. But none of it was connected into a working system. **(deadpan)**"

"What I actually built was the system around them: first, closing the circular loop so that money from selling these products gets reinvested directly back into raw materials and student equipment instead of dying after a one-time exhibition. And second, building the digital storefront so people outside the school could actually see what we make, request samples, and buy from us. That is what turned disconnected classroom activities into EcoLoop. Which brings us to the software." `[NEXT SLIDE]`

`SLIDE 05 • DIGITAL STOREFRONT`

Why Build a Digital Platform?

"Why build a digital storefront for a physical waste project?"

"Because if a circular model only exists inside school walls, it stays a school craft project. Nobody on the outside knows it exists, and you can't generate the revenue to keep the loop reinvesting."

"The website at ecoloop-ssdps.vercel.app unifies the whole model in one public place: it showcases the product catalog, explains how the reinvestment loop works, and gives outside buyers a direct inquiry button to contact us and order physical samples. It bridges the gap between what students make in class and the outside world." `[NEXT SLIDE]`

`SLIDE 06 • $0 BACKEND`

The Zero-Cost Backend: Built on Google Sheets

"Our main constraint on the tech side was simple: we had zero dollars."

"None of us have a company credit card to pay $50 every month for cloud hosting or database instances. So we built a completely free setup:"

"Step 1: The user fills out an inquiry form on the website. Step 2: That data hits a free Google Apps Script webhook in the cloud. Step 3: That webhook logs the order row directly into a Google Sheet, which acts as our database and pings our email. **(pause)** Total monthly running cost: zero dollars. It's free, it works, and it never crashes." `[NEXT SLIDE]`

`SLIDE 07 • ZERO-DOLLAR STACK`

why our database bill is literally $0

"Which brings us to our database bill: literally zero dollars."

"Most people burn fifty dollars a month on cloud database subscriptions just to store basic contact form submissions."

"We run our entire order inquiry backend on a free Google Sheet, and it captures every single lead without costing a penny or needing server maintenance." `[NEXT SLIDE]`

`SLIDE 08 • INQUIRY ENGINE TRUTH`

The Inquiry Engine: An Honest Review

"Now, to be completely honest with you: here's our inquiry modal on the right."

"I built this entire flow in a single night sprint before an exhibition. It is very basic. It pre-populates the product name, checks that the user didn't leave inputs blank, and sends the row to our spreadsheet."

"Is it perfect? Absolutely not. It still needs automated stock deductions, better error states, and tracking numbers. But for an MVP built in one night with zero budget, it solved our immediate problem: taking real inquiries from judges and buyers without spending money." `[NEXT SLIDE]`

`SLIDE 09 • HOW I BUILT IT`

How I Built It Without Knowing React

"Now here's the fun part: how did I actually build this website? Because full disclosure: I don't really know React or TypeScript that deeply."

"The traditional way of coding was brutal. You had to spend six months memorizing syntax, getting stuck on missing brackets, and digging through forum threads because an import broke. Most students give up on their ideas right there."

"The modern shift is: developer as architect. You don't have to memorize boilerplate. You define the system rules, design the user experience, and direct an autonomous AI agent to write the actual code across the files. Your job becomes system design and testing." `[NEXT SLIDE]`

`SLIDE 10 • CHATGPT VS AGENT`

Chatbot vs Agent Workspace

"Here's the difference between using a normal chatbot and using an agent harness."

"On the left is ChatGPT: you ask for code, copy-paste a snippet into your editor, realize it broke something, paste the error back, and after three prompts it forgets your project structure. It has no idea what files exist on your computer."

"On the right is what I actually used: Antigravity, which is an agentic coding environment. An agent has direct access to your repository folder. It reads all your files, edits code across multiple components at once, and runs terminal commands to verify its own work. That's how I was able to build an entire web app without knowing the syntax beforehand." `[NEXT SLIDE]`

`SLIDE 11 • WORKING LOOP`

How Working with an Agent Actually Looks

"This is what that actual day-to-day workflow looks like:"

"Stage 1: You write a spec. Just plain English instructions describing the layout, colors, and behavior."

"Stage 2: The agent edits your files, creating components and routes directly in your folder."

"Stage 3: You test locally. You run the app in your browser, click around, and see what works or what broke."

"Stage 4: You paste the bug back to the agent, and it reads the error and fixes the code. It's a continuous loop of testing and refining." `[NEXT SLIDE]`

`SLIDE 12 • OPENCODE`

Replicating This Workflow: OpenCode

"Now, you don't need access to Antigravity to do this. You can replicate the exact same agent workflow for 100% free using OpenCode."

"OpenCode is an open-source terminal agent built by Anomaly on GitHub at anomalyco/opencode. It lives right inside your terminal. It has a 'Plan Mode' to review your project files without touching them, and a 'Build Mode' that autonomously writes code and runs builds."

"You can install it on any laptop in 60 seconds with: npm install -g @opencode/cli. It runs on Mac, Linux, and Windows." `[NEXT SLIDE]`

`SLIDE 13 • FREE MODELS`

Free AI Models: Zero Cost for Students

"And you definitely don't need to pay for expensive monthly AI subscriptions."

"Option 1: Free Cloud APIs. Google gives you a huge free tier on Google AI Studio for Gemini. Groq Cloud gives you ultra-fast free tokens on Llama 3.3. And GitHub Models lets students use models like GPT-4o right inside VS Code for free."

"Option 2: 100% Offline. You can install Ollama on your laptop, download a model like Qwen 2.5 Coder, and code with an AI agent completely offline with zero internet and zero fees." `[NEXT SLIDE]`

`SLIDE 14 • DEPLOYED WITHOUT TESTING`

deployed without testing

"This was easily the biggest engineering lesson of the entire build. **(deadpan)**"

"The AI will look you dead in the eye and say: 'All bugs resolved, your app is completely ready.' And then you deploy, click the first button on the live site, and the browser console explodes in red errors."

"Prompting is only about 20% of the job. The real 80% is clicking every single button yourself, seeing where the logic broke, and making the agent fix it. Never assume code works until you test it in the browser." `[NEXT SLIDE]`

SLIDE 15 • CONCLUSION & QR

Thank You & Open Discussion

"To wrap this up: EcoLoop shows that high schoolers can tackle real packaging waste and build working software systems for zero dollars using modern open tools."

"If you scan the QR code on the screen, it takes you straight to our GitHub repo at github.com/Harman209/ecoloop. You'll find the full source code, the live website catalog, and a one-page setup guide with exact commands so you can set up OpenCode and free AI models on your own laptop tonight."

"Thank you so much for listening, and I'd love to take your questions." **(conclude recording)**